

INTRODUCTION

- Germany is the world's third-largest economy and the industrial leader of the Eurozone.
- Mittelstand Backbone: Most of German companies are specialized SMEs that drive innovation and provide high employment stability.
- Export-Led Growth: The economy relies heavily on high-end manufacturing, maintaining a massive trade surplus in sectors like automotive and engineering.

This presentation analyses Germany's growth using:

SOLOW GROWTH MODEL

RAMSEY-CASS-KOOPMANS MODEL

**OVERLAPPING GENERATIONS
MODEL**



HISTORICAL BACKGROUND (1945-1990)

Following the devastation of World War II, Germany underwent rapid reconstruction, especially the West

1949

Division into West (market) and East (planned) economies

1950 -1966

Economic miracle (high growth, industrial expansion)

1966 -1967

First recession that ended the Economic Miracle.

1970s

1973 oil crisis leading inflation and slowdown

1980s

Gradual recovery

1990

German reunification

STEPS OF IMPACT

The Reconstruction & SME Support Model (1948)

To provide long-term, low-interest loans for rebuilding war-damaged infrastructure and supporting small-to-medium enterprises

Co-Determination Model (1951)

To give workers a legal voice in company management by allowing them to elect representatives to the board of directors.

The Dual Vocational Training System (1969)

To prevent skill gap by ensuring students receive theoretical classroom knowledge alongside practical, on-the-job training in a real company

Transition from the Deutsche Mark to the Euro (1999)

A shared currency made German exports more price-competitive and stable across the European Union, significantly boosting national trade volumes.

SOLOW GROWTH MODEL

I. Theoretical Context:

Developed independently by Robert Solow and Trevor Swan in 1956, this model provides the primary neoclassical framework for analyzing long-run economic growth. It was formulated as a response to the Harrod-Domar model, seeking to demonstrate that an economy can achieve a stable growth path through flexible factor substitution.

II. Research Objective:

The model investigates how the interaction between capital accumulation, labor force growth, and technological progress determines the long-term trajectory of a nation's real income. It aims to identify the conditions under which an economy reaches a "Steady State", which is, a balanced growth path where the capital-labor ratio remains constant.

III. The Production Function:

The economy is characterized by a neoclassical production function with constant returns to scale (CRS), typically expressed in the Cobb-Douglas form:

$$Y = K^{\alpha} (AL)^{1-\alpha}$$

To facilitate per-capita analysis, the model utilizes the intensive form:

$$y = f(k)$$

where y is output per effective worker and k is capital per effective worker. This highlights the principle of diminishing marginal productivity: as capital intensity increases, the incremental gain in output decreases.

IV. The Dynamics of Capital Accumulation:

The evolution of the economy is governed by the fundamental law of motion for capital:

$$\dot{k} = sf(k) - (n + g + \delta)k$$

Here, $sf(k)$ represents actual investment per effective worker, and, $(n + g + \delta)k$ represents the “break-even” investment, that is, the amount required to maintain the capital-labor ratio against the pressures of: population growth (n), technological advancement (g), and the depreciation of physical capital (δ).

V. The Steady-State Equilibrium:

The central prediction of Solow-Swan model is the existence of a **steady state** (k^*), where investment exactly offsets capital dilution ($\dot{k} = 0$).

In this equilibrium:

- Output per worker grows only at the rate of technological progress (g).
- Differences in the savings rate (s) explain differences in the level of income, but not the long-run growth rate.

BARROW REGRESSION ANALYSIS

- **Period 1: 1962–1991**
- Post World War II reconstruction
- Growth driven by:
 - Capital accumulation
 - Industrial expansion

$$g_t = \beta_0 + \beta_1 \ln(Y_{t-1}) + \beta_2 Inv_{t-1} + \beta_3 Pop_{t-1} + \epsilon_t$$

The model tests Solow predictions on growth, convergence, and factor accumulation using lagged macroeconomic variables

Period 2: 1992–2024

- After German reunification
- Growth driven by:
 - Technology
 - Human capital

$$g_t = \beta_0 + \beta_1 \ln(Y_{t-1}) + \beta_2 Inv_{t-1} + \beta_3 \ln(HC_{t-1}) + \beta_4 Pop_{t-1} + \epsilon_t$$

THE REGRESSION OUTPUT & CONDITIONAL CONVERGENCE

Model 1: OLS, using observations 2-30 (n = 29)

Dependent variable: GrowthRateg

Heteroskedasticity-robust standard errors, variant HC1

	coefficient	std. error	t-ratio	p-value

const	30.9070	19.5027	1.585	0.1256
lnYt_1	-0.919702	0.667449	-1.378	0.1804
Invt_1	-0.124740	0.121384	-1.028	0.3140
Popt_1	0.591326	1.25275	0.4720	0.6410
Mean dependent var	3.186839	S.D. dependent var	2.102838	
Sum squared resid	104.5820	S.E. of regression	2.045307	
R-squared	0.155330	Adjusted R-squared	0.053969	
F(3, 25)	1.631752	P-value(F)	0.207243	
Log-likelihood	-59.74801	Akaike criterion	127.4960	
Schwarz criterion	132.9652	Hannan-Quinn	129.2089	

Excluding the constant, p-value was highest for variable 8 (Pop

Model 1: OLS, using observations 1992-2024 (T = 33)

Dependent variable: GrowthRate

HAC standard errors, bandwidth 2, Bartlett kernel

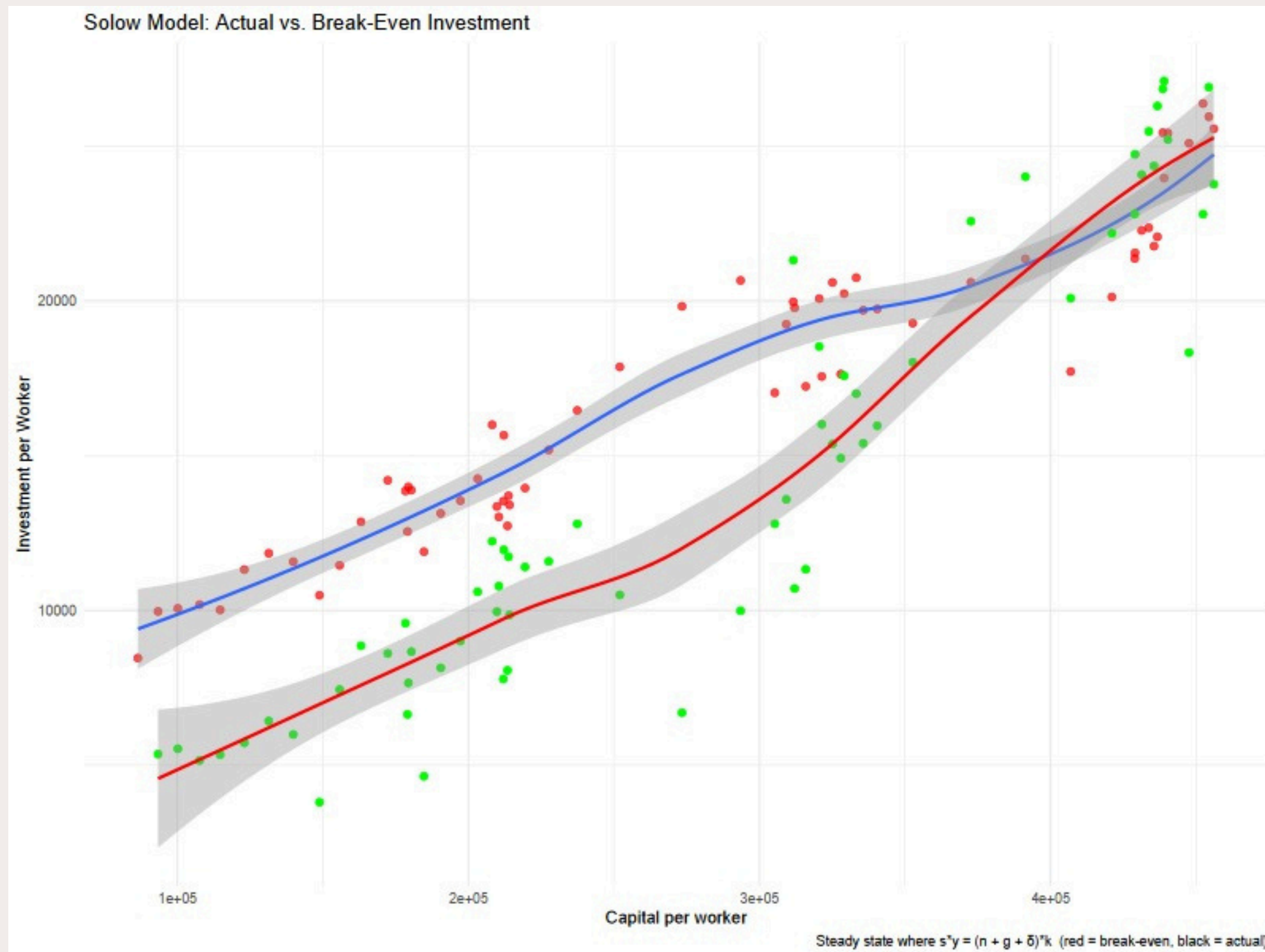
	coefficient	std. error	t-ratio	p-value

const	41.7420	46.2962	0.9016	0.3749
lnYt_1	-2.18905	2.99123	-0.7318	0.4704
Invt_1	0.000683080	0.0962222	0.007099	0.9944
Popt_1	0.227804	0.482181	0.4724	0.6403
lnHCt_1	17.4140	34.1375	0.5101	0.6140
Mean dependent var	1.215541	S.D. dependent var	2.074386	
Sum squared resid	133.4926	S.E. of regression	2.183482	
R-squared	0.030545	Adjusted R-squared	-0.107949	
F(4, 28)	0.403919	P-value(F)	0.804174	
Log-likelihood	-69.88435	Akaike criterion	149.7687	
Schwarz criterion	157.2512	Hannan-Quinn	152.2864	
rho	-0.097842	Durbin-Watson	2.179582	

Excluding the constant, p-value was highest for variable 6 (Invt_1)

The regression results provide weak empirical support for the Solow model in explaining Germany's growth. Although the sign of initial income suggests conditional convergence, the coefficients are statistically insignificant. This indicates that Germany's growth was influenced by broader structural and technological factors beyond the basic model.

Steady-State Dynamics in the Solow Model



At k^* :

- $sY = (n + g + \delta)k$
- So,
- Net investment = 0

Meaning:?

- Capital is not increasing
- Capital is not decreasing
- It is constant over time

RAMSEY–CASS–KOOPMANS MODEL

- Germany's post-reunification period involved capital accumulation and structural adjustment, aligning with RCK dynamics.
- The economy shows stable long-run growth, moderate population growth, and steady technological progress.
- High savings and investment behavior reflect forward-looking consumption decisions, consistent with the model.
- These features make Germany suitable for analyzing transition to steady state and optimal capital accumulation.



Model 1: OLS, using observations 1991–2023 (T = 33)
Dependent variable: logGDP

	coefficient	std. error	t-ratio	p-value	
const	-21.9467	8.27144	-2.653	0.0126	**
logL	2.75802	0.546668	5.045	2.05e-05	***
logK	0.140439	0.0845102	1.662	0.1070	

Mean dependent var
Sum squared resid
R-squared
F(2, 30)
Log-likelihood
Schwarz criterion
rho

28.75645
0.031746
0.938548
229.0935
67.79214
-125.0948
0.675527

S.D. dependent var
S.E. of regression
Adjusted R-squared
P-value(F)
Akaike criterion
Hannan-Quinn
Durbin-Watson

0.127058
0.032530
0.934451
6.73e-19
-129.5843
-128.0737
0.545175

A Cobb–Douglas production function was estimated using OLS in Gretl to understand the empirical relationship between output, capital, and labour:

$$\ln Y = \beta_0 + \alpha \ln K + \beta \ln L$$

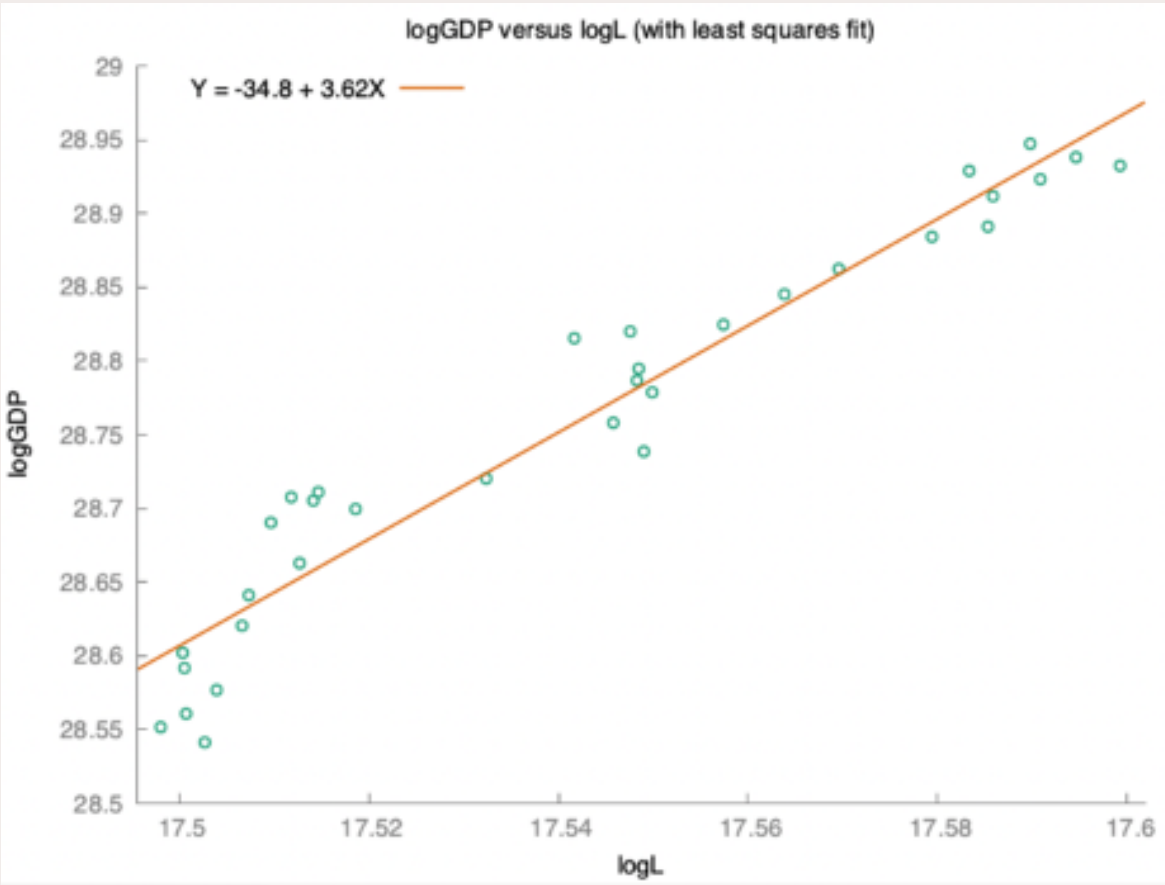
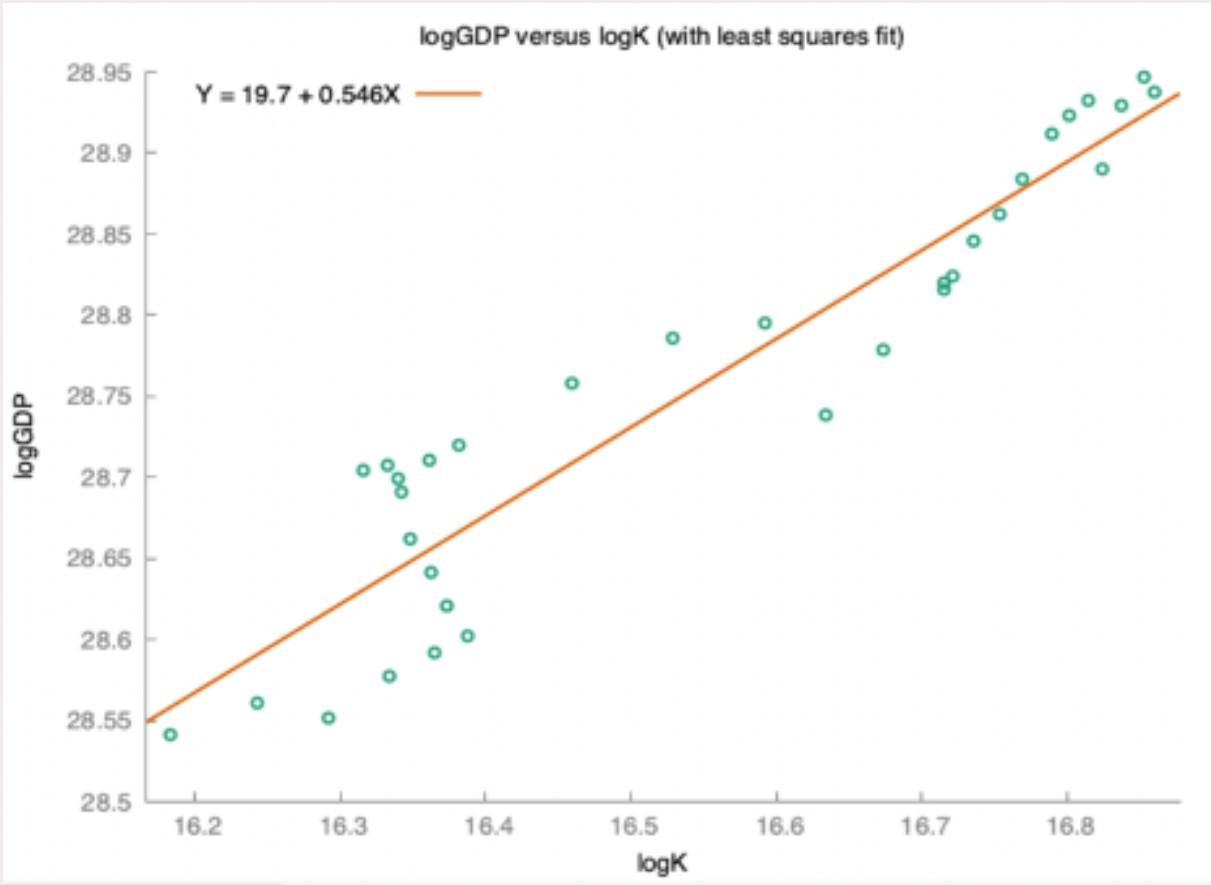
The regression provides following input:

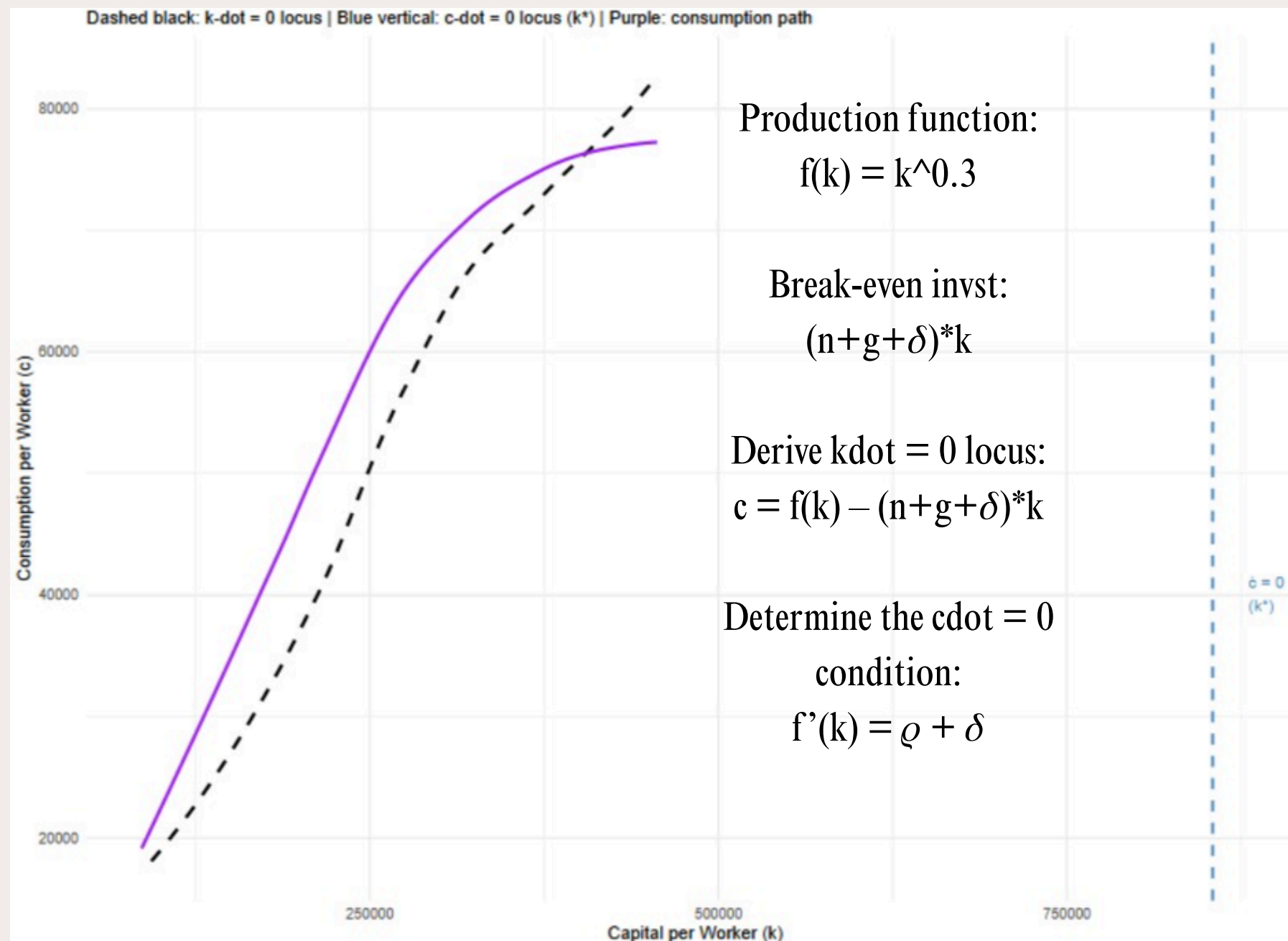
- Labour has a very strong and statistically significant effect on output.
- Capital accumulation appears to have a positive but relatively small effect on GDP.
- $R^2 = 0.938$, means 93.8% of the variation in German GDP is explained by labour and capital.

The core equations of the model that are derived from the optimization behaviour of a representative household that maximizes lifetime utility subject to a budget constraint.are:

$$\dot{k} = f(k) - c - (n+g+\delta)k$$

$$\dot{c}/c = 1/\theta [f'(k) - \rho - \delta]$$





$$\delta=0.05$$

$$g=0.02$$

$$\alpha=0.33$$

Steady state occurs where:
 $\dot{k}=0$ and $\dot{c}=0$

This gives optimal capital and consumption.

Empirical interpretation

The diagram suggests:

- Germany is moving toward steady state
 - Consumption adjusts gradually
 - Capital accumulation remains stable
- Transition dynamics dominate short run movements

The graph captures the resource constraint of the economy but does not fully represent the Ramsey equilibrium, as it lacks the consumption dynamics and saddle path. This limitation arises partly due to the use of limited data and small values of capital, which restrict the ability to accurately construct the full phase diagram. Since the Ramsey model is highly sensitive to parameter values and functional forms, using a finite dataset and approximated capital values makes it difficult to derive the precise steady-state and dynamic adjustment path.

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Function evaluations: 52
Evaluations of gradient: 10

Model 1: 2-step GMM, using observations 1972-2022 (T = 51)
Dependent variable: lnconsgrowth
Instrumented: RIR_1
Instruments: const lnconsgrowth_1 lnconsgrowth_2 RIR_2
HAC standard errors, bandwidth 2, Bartlett kernel

      coefficient    std. error      z      p-value
-----
const      0.0172280    0.00269987   6.381   1.76e-010 ***
RIR_1      0.0373469    0.0116784   3.198   0.0014 ***

Mean dependent var    0.014743    S.D. dependent var    0.018569

GMM criterion: Q = 0.0300196 (TQ = 1.531)
J test: Chi-square(2) = 1.531 [0.4651]

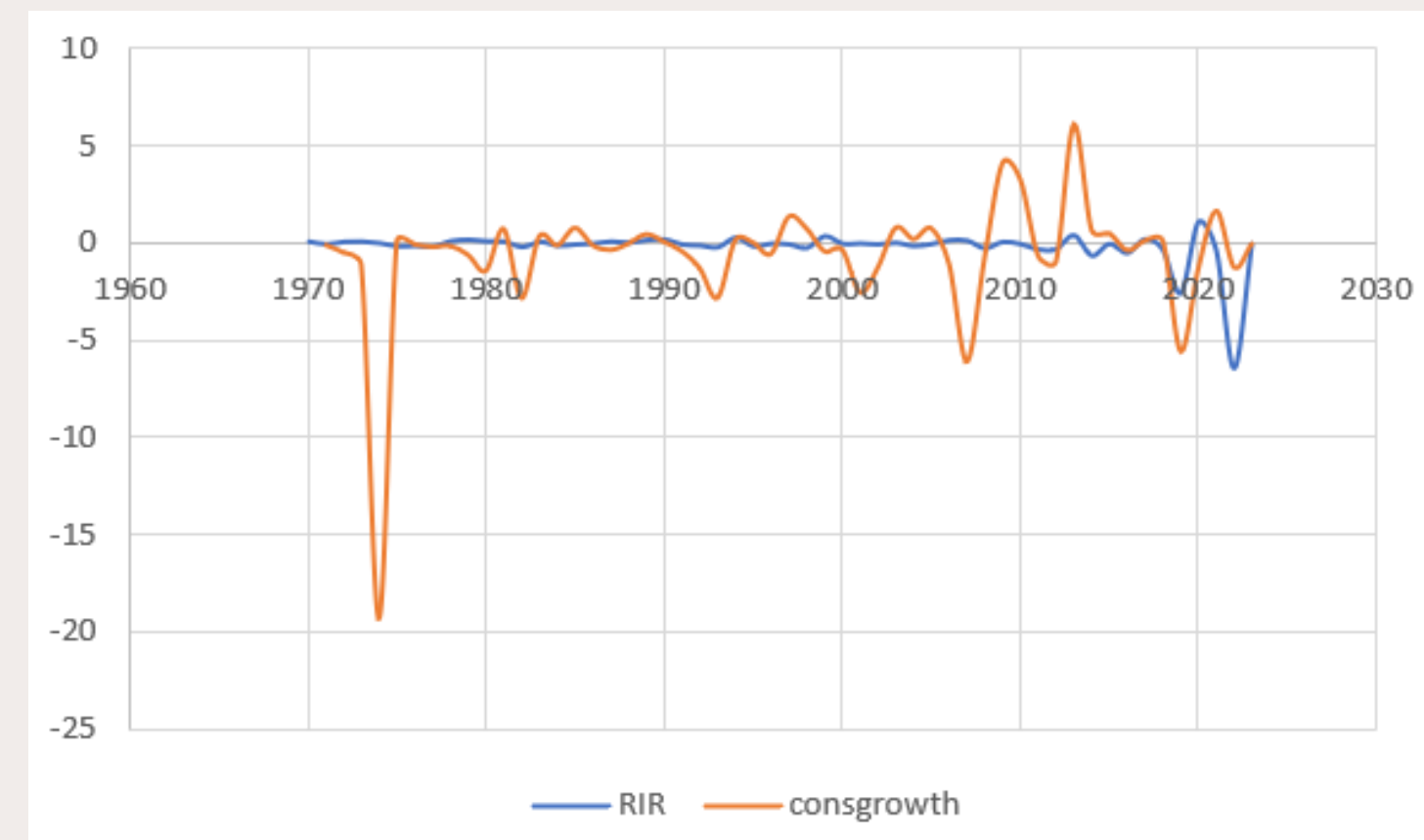
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- Employing GMM instead of OLS to estimate the Euler equation addresses endogeneity in the real interest rate and expectations.
- Following “Empirical evidence on the Euler equation for consumption in the US (2021)”, we log-linearise the Euler equation, obtaining:

$$\Delta \ln C_t = \alpha + \beta r_{t-1} + \varepsilon_t$$

- Coefficient on lagged real interest rate is 0.037 (significant). A 1 unit increase in the real interest rate raises consumption growth by 0.037 percentage points.
- The sign is consistent with theory, but the magnitude is small, indicating low intertemporal substitution (IES).
- Intercept 0.017 captures average consumption growth, not purely the structural preference parameter.

- Consumption growth is more volatile than real interest rates because it is a change variable and responds to multiple economic shocks such as income fluctuations, expectations, and crises.
- Interest rates are smoother due to monetary policy smoothing.
- Weak response of consumption to interest rates, consistent with low IES, as reflected in the small GMM coefficient
- This implies that households are reluctant toward large shifts in consumption and exhibit a strong tendency to smooth consumption over time.



OVERLAPPING GENERATIONS MODEL

The OLG model analyzes an economy where individuals live for two periods (young and old), allowing for explicit study of intergenerational interactions, savings behavior, and capital accumulation.

1. Basic Structure

- Individuals are young in period t and old in period $t+1$
- Young: earn income, consume, and save
- Old: consume from past savings (no labor income)
- Savings of the young become capital stock in the next period

2. Dynamic Equilibrium

- Economic outcomes depend on intertemporal optimization
- Capital evolves according to aggregate savings decisions
- The economy may converge to a steady-state equilibrium, conditional on initial capital stock

. Dynamic Inefficiency

- Occurs when capital exceeds the Golden Rule level
- Condition: Marginal Product of Capital (MPK) $<$ population growth rate (n)
- Indicates over-accumulation of capital (over-saving)
- Implies scope for Pareto improvement (e.g., reducing savings or introducing public debt)

4. Transition Dynamics

- Describes how the economy adjusts from one steady state to another
- Triggered by policy changes, shocks, or structural shifts
- Adjustment occurs through changes in:

1. Capital stock (K)
2. Interest rate (r)
3. Savings behavior over time

Data and Interpretation

Dataset Overview :- The dataset covers the period 1976-2024 and combines key macroeconomic variables

GDP Growth(g)

- Represents the growth in output and productivity in the economy

Population Growth(n)

- It influences labour supply, savings behaviour and capital accumulation

CPI(Consumer Price Index) π

- It shows the changes in the price level over time (Inflation)

Nominal Interest Rate(i)

- It is shown in the form of 10 year government bond yield

Real Interest Rate (r)

- It is calculated using the Fisher equation $RIR (r) = i - \pi$

Combined Growth Rate($n+g$)

Represents total economic growth by combining population and productivity growth

Average RIR(r^*) / Steady State

It reflects the real reward for saving and investing in financial assets. It is the short run value of r .

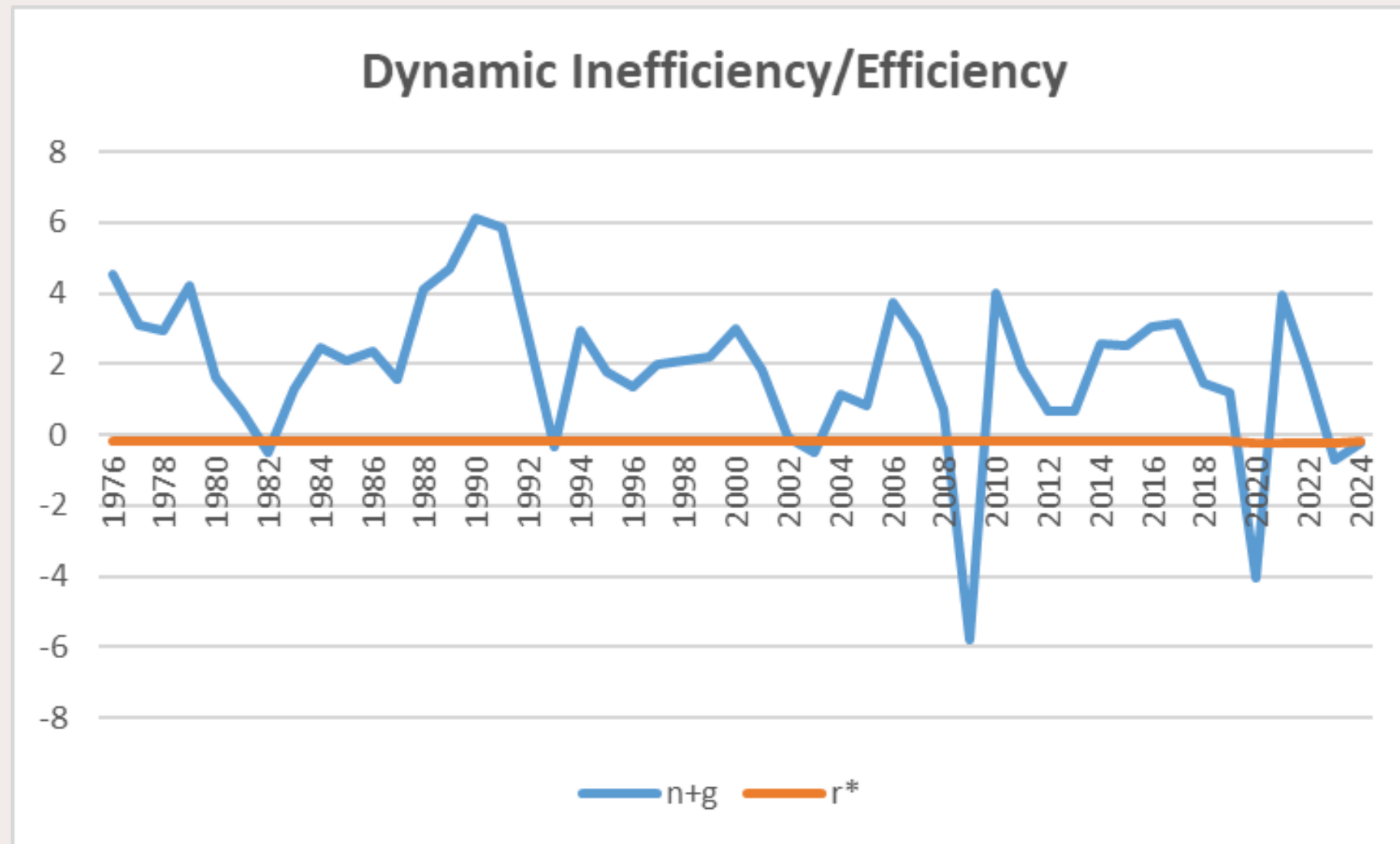
Gross savings

It represents the portion of income not consumed and is obtained from the World Bank.

Capital Transition Equation

It shows how current savings become future capital. $K_{t-1} = s/100 + n$

Dynamic Inefficiency/Efficiency



Dynamic Inefficiency

- Dynamic Inefficiency occurs when the return on capital is lower than the growth rate of the economy.
- $r^* < n+g$
- A Dummy variable is created to find out Dynamic Inefficiency/ Efficiency. In excel we have used the IF formula to come to this result
- $DI = 1$ when $r^* < n+g$ (Dynamic Inefficiency exists)
- $DI = 0$ when $r^* > n+g$ (Dynamic efficiency exists)

Reasons :-

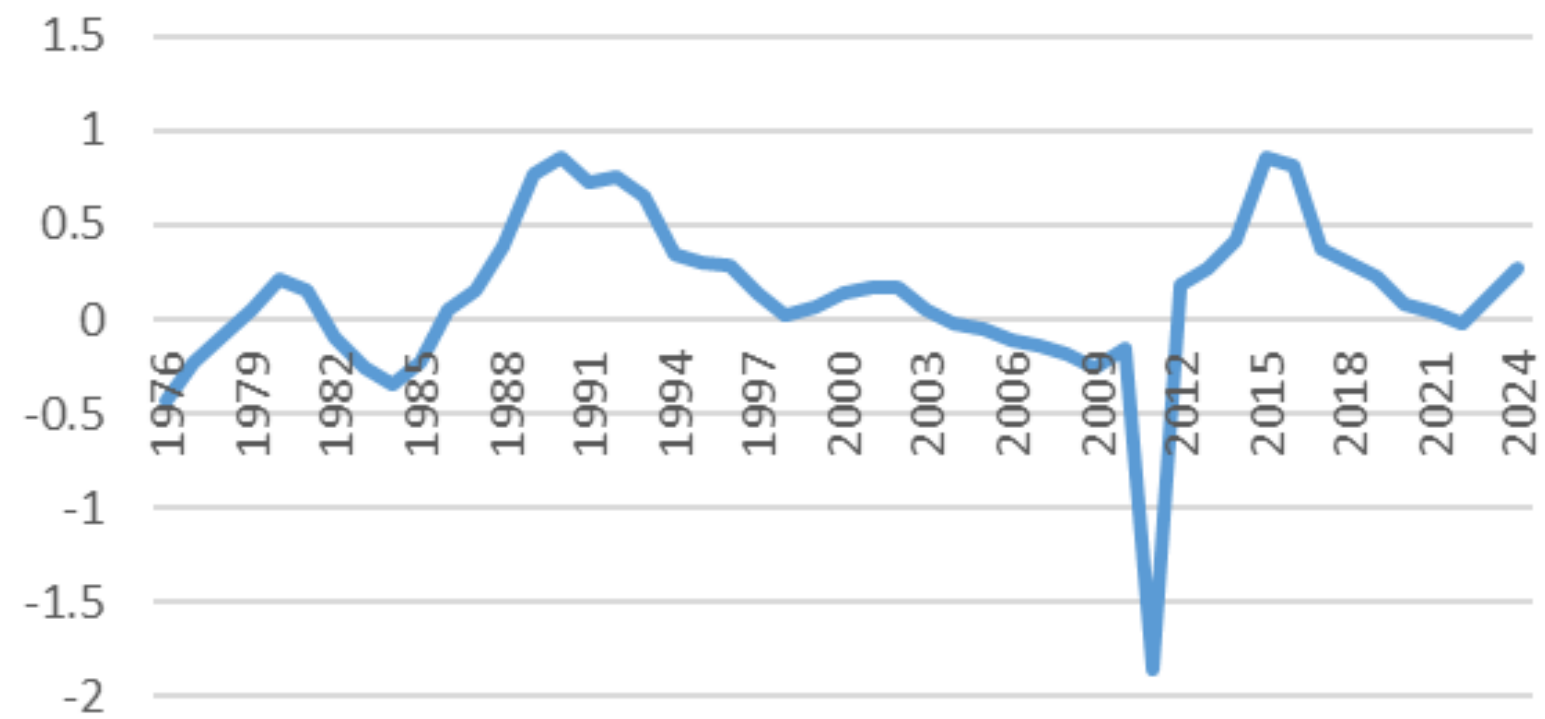
Dynamic Inefficiency ($r^* < n + g$)

- Low real interest rates ($r^* \downarrow$)
- High savings (aging population)
- Excess capital accumulation
- Monetary easing by European Central Bank
- Strong economic growth ($n + g \uparrow$)

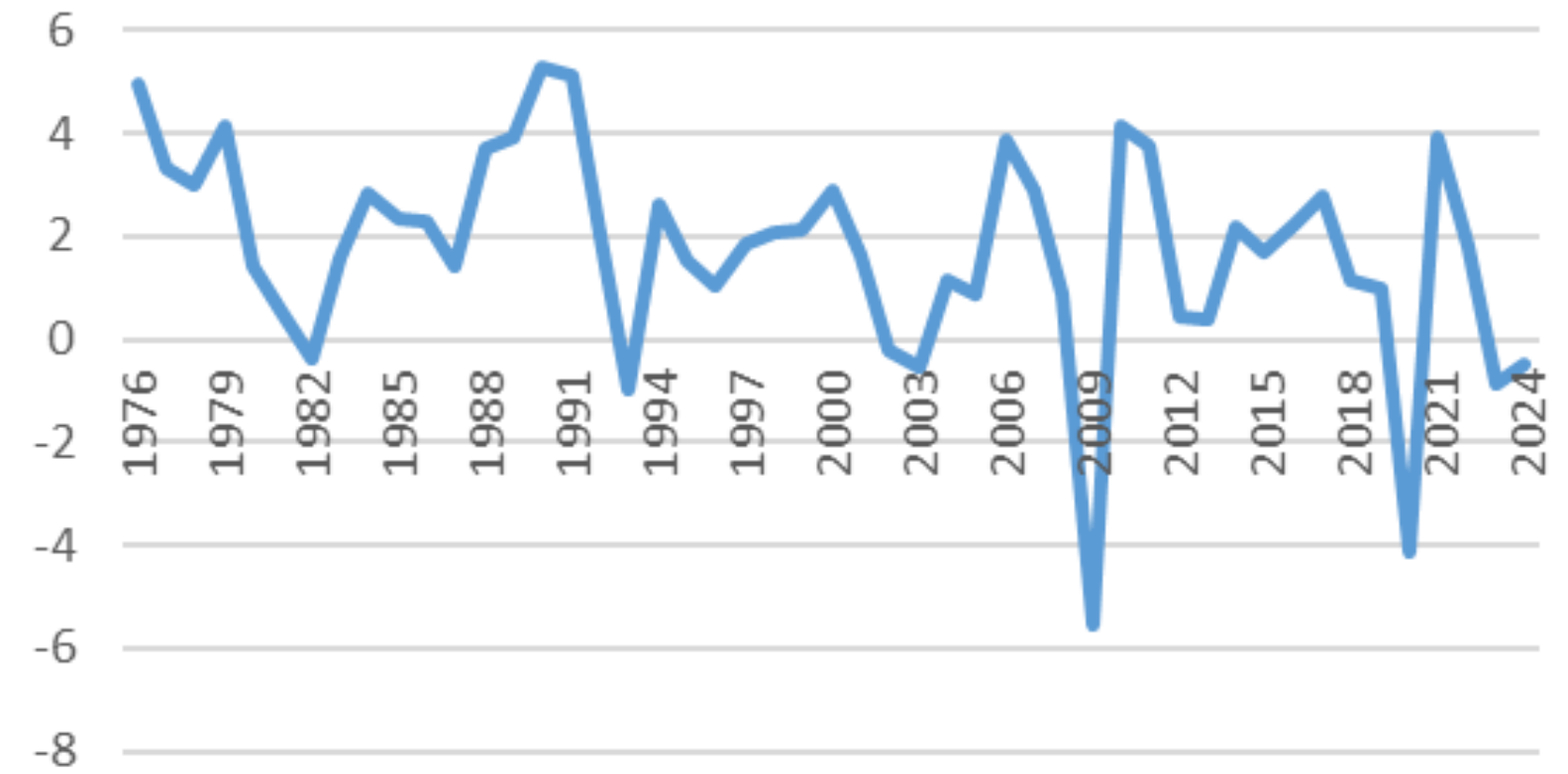
Dynamic Efficiency ($r^* > n + g$)

- Higher real interest rates ($r^* \uparrow$)
- Slower population & productivity growth
- Capital scarcity after shocks
- Financial market corrections

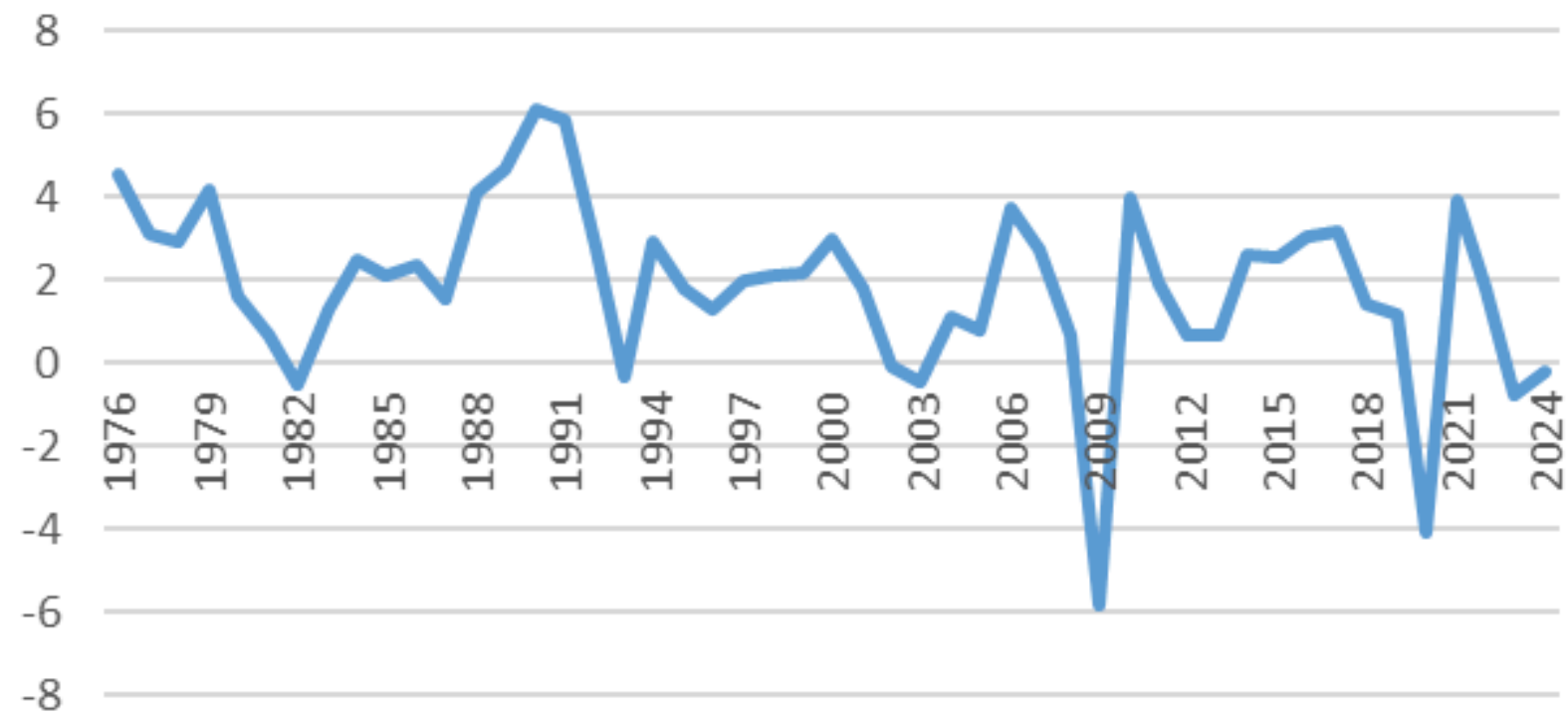
Population growth



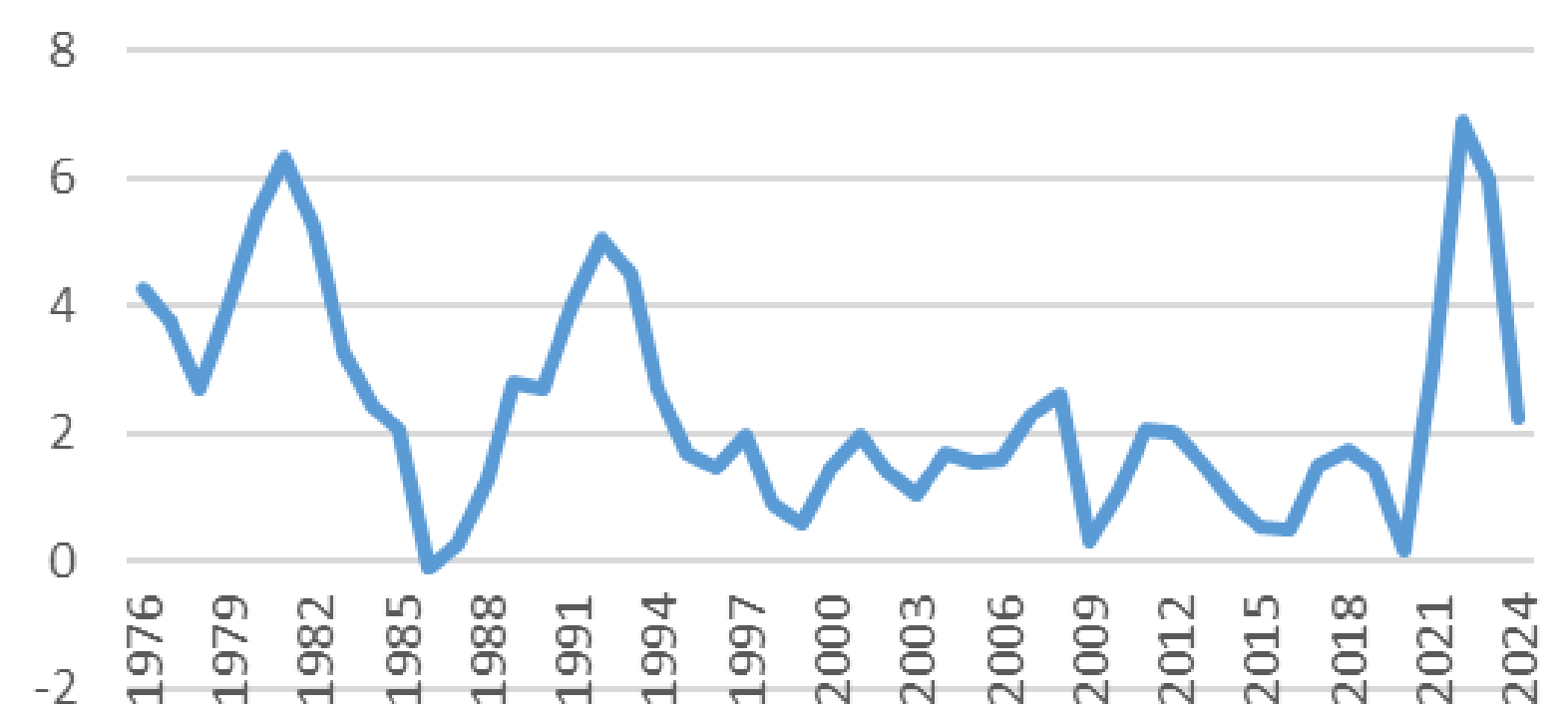
GDP growth



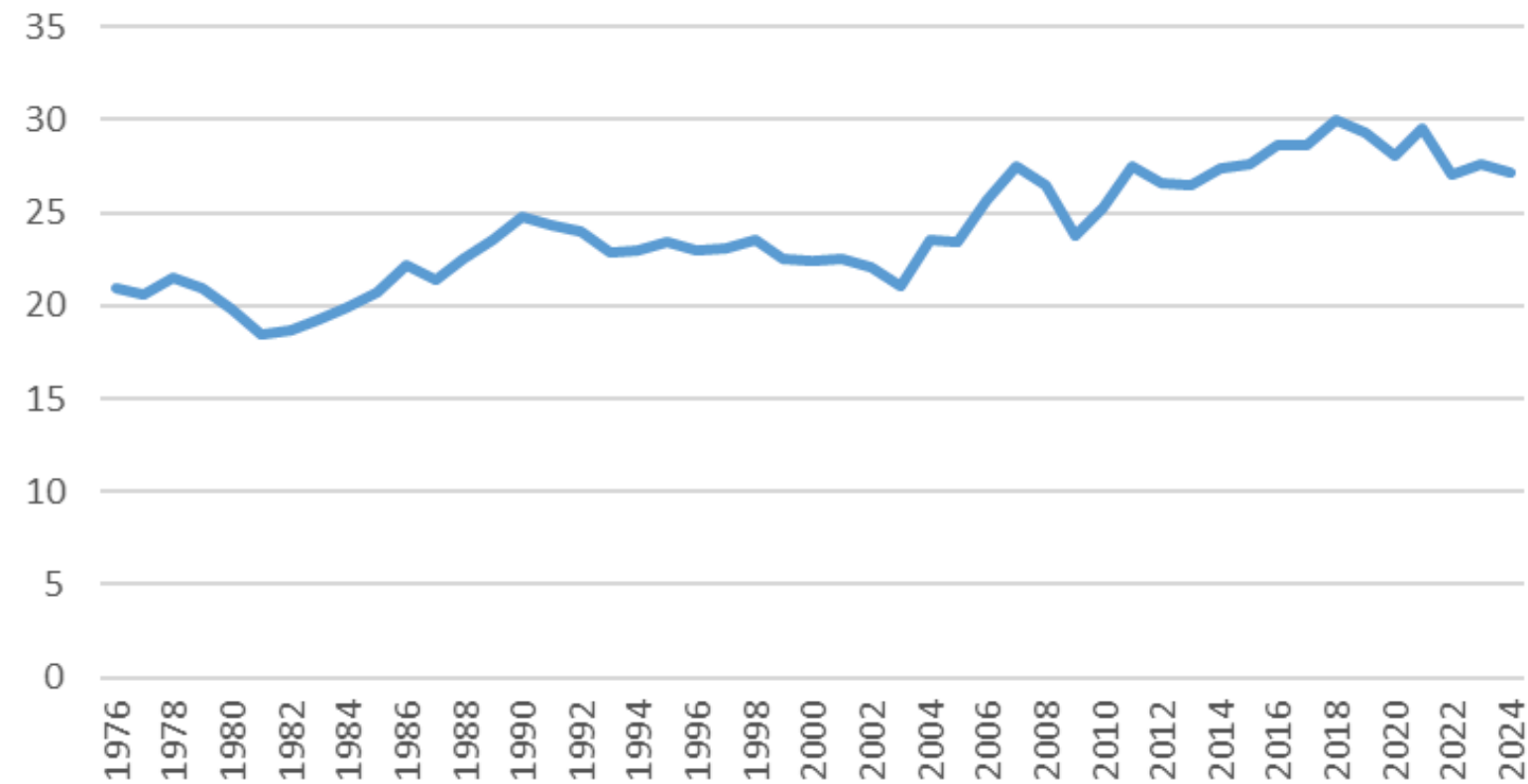
n+g



CPI



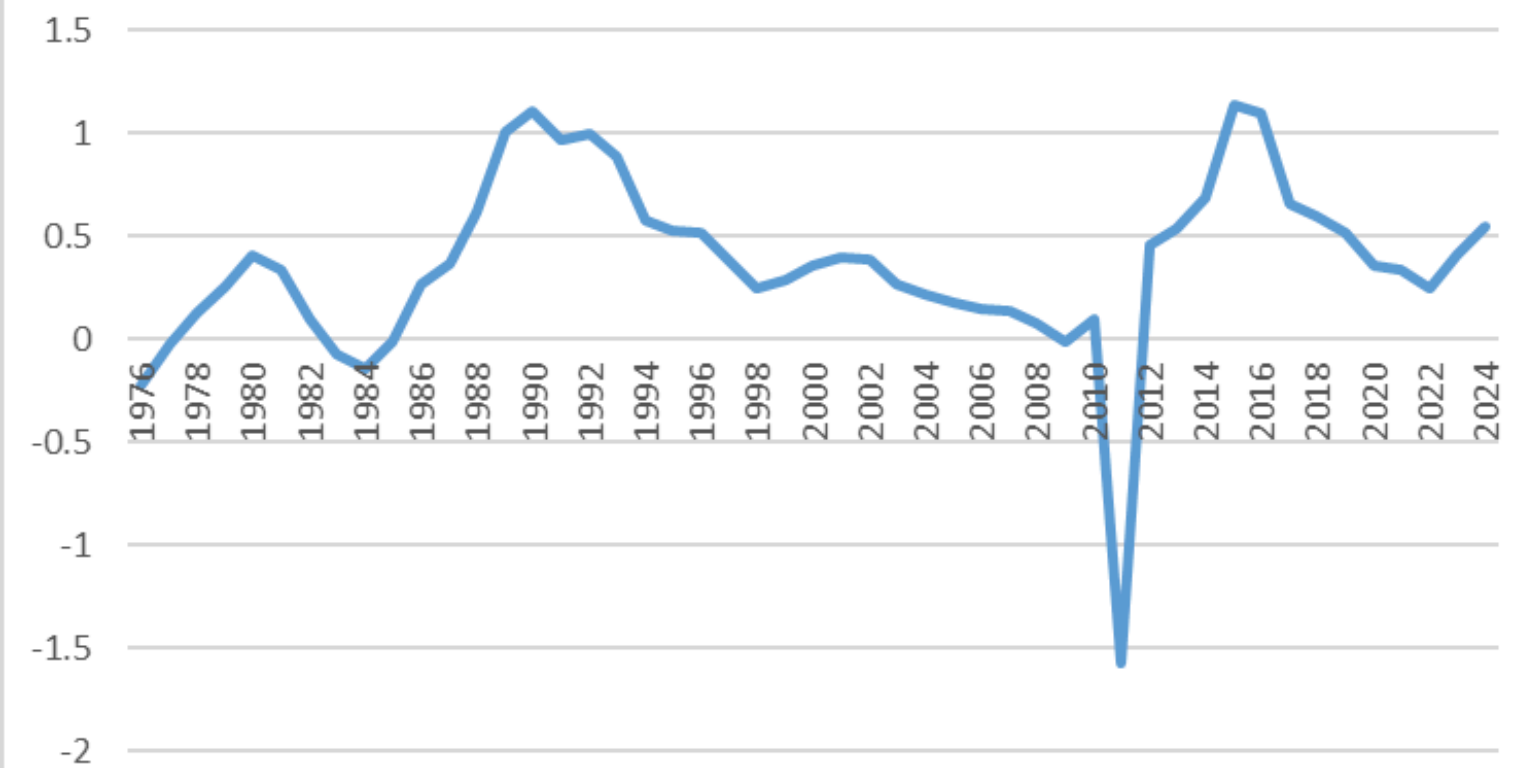
Savings(s)



Germany's savings trend shows generally stable but gradually increasing behaviour over time, with noticeable rises during periods of uncertainty. Savings remained steady in earlier decades, increased after the early 2000s, and rose sharply during crises like 2008 and especially in 2020 due to restricted spending. In recent years, savings have slightly declined but remain relatively high.

Overall, Germany has a strong saving culture, with households saving more during uncertain times and maintaining consistently high saving levels.

Germany's capital formation behaves like a cycle of rise and fall, depending on economic conditions. For example, investment increased sharply during the late 1980s–early 1990s with economic expansion, but dropped significantly during the 2008–2009 financial crisis when uncertainty was high. It later recovered as the economy improved. In recent years, capital growth has been more stable but moderate, showing balanced investment and steady economic conditions.

Capital Transition($kt+1$)

Model Comparison

Analytical and Empirical Evaluation Model Comparison: Solow vs Ramsey vs OLG

Solow Model

Prediction:

- Growth driven by capital, labour and technology
- Conditional convergence expected

Evidence:

- Convergence is seen.
- Investment not strongly significant
- Structural factors more important

Inference:

Explains long-run structure but ignores behavioural factors.

Ramsey Model

Prediction:

- Growth depends on optimal consumption and savings
- Interest rates affect consumption growth

Evidence:

- Labour strongly significant
- High explanatory power ($\text{Adj } R^2 \approx 0.93$)
- Weak response of consumption to interest rate

Inference:

Better explains stability and forward-looking behaviour.

OLG Model

Prediction:

- Demographics affect savings and capital
- Dynamic efficiency depends on r vs $(n+g)$

Evidence:

- Aging population effects
- Strong savings behaviour
- Periods of $r < n+g$

Inference:

Explains demographic and financial structure.

Germany's growth is better explained by Ramsey and OLG because growth depends on: Human capital, Demographic transition, Savings behaviour & Institutional strength rather than only capital accumulation.

Policy Implications: Germany's Growth Strategy

1. Productivity & Growth Policies

(Evidence: Labour strongly affects GDP)

Policies:

- Invest in human capital and skills
- Strengthen vocational training systems
- Support technology adoption in SMEs

2. Savings & Financial Stability Policies

(Evidence: Strong savings behaviour)

Policies:

- Encourage productive investment channels
- Strengthen financial market stability
- Promote innovation financing

3. Demographic Policies

(Evidence: Aging population & OLG results)

Policies:

- Pension system reforms
- Encourage skilled immigration
- Increase female labour participation
- Support family welfare policies

4. Macroeconomic Stability Policies

(Evidence: Consumption smoothing behaviour)

Policies:

- Maintain stable inflation
- Predictable monetary policy
- Reduce economic uncertainty

5. Structural Growth Policies

Policies:

- Innovation driven growth
- Labour productivity
- Demographic sustainability
- Long-term competitiveness

Conclusion

Germany's growth shows a shift from capital-driven (post-WWII) → human capital & technology

Solow Model

- Weak empirical support (low $R^2 \approx 0.15$, insignificant coefficients)
- Explains early growth, but fails for modern Germany

Ramsey Model Strong fit ($R^2 \approx 0.938$)

- Labour highly significant; capital less impactful
- Consumption responds weakly to interest rates ($\beta \approx 0.037 \rightarrow$ low IES)

OLG Model Highlights ageing population & high savings behaviour

- Evidence of dynamic inefficiency ($r < n+g$) in periods
- Savings rise during crises (2008, 2020)

Criticism of Models

Solow Model

Poor empirical performance (low R^2 , insignificant results)

Overly simplistic $\rightarrow$ reduces growth to capital accumulation

Cannot explain Germany's post-reunification, innovation-led growth

Ramsey Model

Empirically weak behavioural channel (very low $\beta = 0.037$)

Results depend heavily on assumed parameters such as δ

Fails to fully capture real dynamics due to data and modelling constraints

OLG Model

Relies on stylised assumptions (2-period life, representative agents)

Dynamic inefficiency results are sensitive to the measurement of r , n , g

Does not incorporate institutional and policy complexities